Dhairya Thakker

Panvel, India | +91 96192 97756 | dhairyathakker76@gmail.com | linkedin.com/in/thakker-dhairya | github.com/Dhairya76

Summary

Backend-focused Software Engineer with 3+ years architecting and scaling mission-critical enterprise KYC, identity verification, and anti-fraud platforms for tier-1 banks and fintechs. Hands-on depth across Node.js/Fastify microservices, distributed data streaming, edge ML/LLM pipelines, and enterprise security hardening (E2E encryption, VAPT remediation, zero-trust RBAC). Proven record of reducing database footprint by ∼32%, mitigating OWASP attack surfaces, and maintaining zero-downtime microservices on GCP and Kubernetes.

Technical Skills

Languages: JavaScript (ES6+), TypeScript, Python, SQL
Backend & Distributed Systems: Node.js, Fastify, REST APIs, WebSockets, Sequelize, RabbitMQ, Redis, Google Cloud Pub/Sub
Databases & Analytics: PostgreSQL, ClickHouse, BigQuery, Redis, Logstash, Apache Beam, Metabase, Grafana
AI, Edge ML & Security: OpenAI & Gemini APIs, ONNX Runtime Web (RF-DETR), VAPT, AES/A128CBC, HMAC, RBAC
Cloud & Infrastructure: GCP (GCS), Kubernetes, Docker, ArgoCD, GitLab CI/CD, Git

Work Experience

Algorisys Technologies (Client Deputation: IDfy) — Mumbai, India
Software Engineer | Enterprise KYC & Identity Platform — *Sep 2023 – Present*

- **Distributed Backend & Async Ingestion:** Architected a dedicated reporting microservice offloading heavy CSV/Excel processing from the core API, streaming batched data exports directly to GCS with configurable column filters and scheduled email dispatch; built streaming sheet-ingestion APIs handling high-volume UUID records.
- **Event-Driven Comms & Notifications:** Engineered an end-to-end event notification engine and reviewer console tray backed by Google Cloud Pub/Sub, implementing conditional retrigger mechanisms and stateful alert queues for enterprise verifier workflows.
- **Production AI & Edge Vision Integration:** Integrated client-side RF-DETR object detection models using **ONNX Runtime Web** for automated, real-time ID document and selfie auto-capture; deployed resilient LLM name-parsing workflows (**OpenAI API with automatic Gemini API fallback**).
- **End-to-End Cryptography & Encryption:** Designed per-session WebSocket payload encryption using A128CBC with client-level feature-flag toggling and fail-closed security guarantees, ensuring strict non-tampering of in-flight journey records.
- **VAPT Hardening & OWASP Remediation:** Remediated critical/high banking security vulnerabilities: eliminated null-byte extension bypasses, blocked malicious PDF/CSV injection vectors, resolved SQLi, and added HMAC-timestamped webhooks, OTP rate limiting, and Cloudflare Turnstile CAPTCHA.
- **Access Control & Governance:** Implemented server-side Role-Based Access Control (RBAC) across decisioning APIs (approval/rejection audit trails) and token refresh lifecycles, satisfying strict multi-tenant banking compliance.
- **Data Lifecycle & Storage Optimization:** Shipped an OU-scoped data purge and archival engine with dynamic TTL retention; eliminated worker OOM crashes via serialized batch processing and reclaimed **∼32% database storage (22 GB → 15 GB)** using scheduled `VACUUM FULL / ANALYZE`.
- **Streaming Telemetry & Analytics:** Deployed Logstash ingestion parsers feeding ClickHouse and BigQuery schemas with automated partition TTLs; tracked step turnaround times (TAT), retry counts, and funnel drop-off analytics feeding real-time Metabase dashboards.
- **International Fintech Rollouts:** Shipped compliant KYC/KYB journeys across 4 global markets (India, Philippines, UAE, Indonesia), handling 17+ document types, MRZ checksum validation, 4K camera capture, and National-ID QR parsing with Latin character support.

Education

Shah & Anchor Kutchhi Engineering College (University of Mumbai) — Mumbai, India
Bachelor of Engineering (B.E.) in Information Technology — *2019 – 2023*